KEONSOO LEE

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EDUCATION

KOREA AEROSPACE UNIVERSITY, GOYANG, SOUTH KOREA 03/2021–08/2027 (Expected)
BACHELOR OF SCIENCE (B.S.) IN SOFTWARE ENGINEERING
GPA: 3.70/4.50
Selected Coursework: Probability and Statistics, Deep Learning, Generative AI Applications, Analysis and Design of Algorithms, Data Structures, Computer Architecture, AI Programming.

UNIVERSITY OF NEVADA, LAS VEGAS, USA Spring 2026
Non-degree Coursework: Introduction to Machine Learning (3 credits; Grade: A−)

RESEARCH INTERESTS

- Small Language Models
- Efficient Training and Inference
- Language Model Architectures

RESEARCH INTERNSHIPS

QUANTUM ARTIFICIAL INTELLIGENCE (QAI) LAB, KAU 06/2025–Present
UNDERGRADUATE RESEARCHER | ADVISOR: PROF. JAY HOON JUNG

DISC LAB, UNIVERSITY OF NEVADA, LAS VEGAS, USA 01/2026–05/2026
RESEARCH INTERN | ADVISOR: PROF. SHAIKH ARIFUZZAMAN

RESEARCH EXPERIENCE

LANGUAGE-MODEL TRAINING DATA CURATION 01/2026–08/2026
FIRST AUTHOR | DISC LAB, UNLV

- Led framework design, implementation, experiments, and manuscript writing for auditable language-model data curation.
- Integrated four core concepts—Validity, Redundancy, Quality, and Coverage—to govern data selection and preserve traceable membership decisions.
- Evaluated Qwen3-4B-Base via QLoRA continued pretraining across three seeds and six code benchmarks; reduced corpus tokens by 27.95%, with a mean primary reasoning macro score of 20.80 versus 20.86 for Raw.

OPTIMABS: EFFICIENT ADAPTIVE OPTIMIZATION 05/2025–09/2025
FIRST AUTHOR AND PRESENTER | QAI LAB

- Proposed and implemented adaptive optimizer variants using absolute-gradient accumulation to eliminate square-root operations; compared them with RMSProp, Adam, and AdamW on a fixed MLP architecture.
- On MNIST with batch size 64, reduced training time by 5.7% versus RMSProp and 3.7% versus Adam, with test-accuracy decreases of 0.68 and 0.78 percentage points, respectively.
- Wrote a substantial portion of the manuscript.

APPLE: Attention with a Low-Rank Residual Path 09/2026

Second Author | QAI Lab

- Contributed to manuscript writing and ran language-model benchmarks on Pythia-based models incorporating the proposed architecture.

AI Trustworthiness for Aviation MRO 09/2025–12/2025

Undergraduate Researcher | QAI Lab

- Reviewed EASA documents and research papers on AI trustworthiness for aviation maintenance, repair, and overhaul (MRO); organized and shared summaries with the research team.

AI-Generated Korean Text Detection 07/2025

Team Member | SW-Centered University Digital Competition

- Fine-tuned KoBERT to distinguish human-written and AI-generated Korean text.
- Explored an XGBoost ensemble combining BERT embeddings with stylometric features.

PUBLICATIONS AND MANUSCRIPTS

- **Keonsoo Lee**, Jeong Hun Kim, Ji Hong Kim, and Jay Hoon Jung. "OptimABS: Computational Efficiency Improvement of Absolute Value-based Adaptive Optimizer." *Proceedings of the 6th Korea Artificial Intelligence Conference (KAIC 2025)*, September 2025, pp. 358–359.
- **Keonsoo Lee** and Shaikh Arifuzzaman. "Evidence-Bound Curation: Auditable Membership Decisions for Language-Model Pretraining Data." *Submitted to IEEE Big Data 2026, August 21, 2026.*
- "APPLE: Recovering Attention-Attenuated Evidence with a Low-Rank Residual Path." *Submitted to ICLR 2027 (second author).*

AWARDS AND HONORS

- Overseas Study Scholarship, Korea Aerospace University (KRW 4,800,000; Spring 2026), supporting the UNLV research internship.
- Scholarship for Academic Excellence (2024).

OTHERS

Language Proficiency

- English: TOEIC 825 | OPIc AL

Teaching and Leadership

- Teaching Assistant, Korea Aerospace University: Data Structures (Spring 2025) and AI Programming (Fall 2025).
- President, Deeperent AI Study Group (July 2026–Present).